

EMOTIONAL FIDELITY IN AUDIO LANGUAGE MODELS: BASELINE ANALYSIS

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ABSTRACT

This report investigates emotional fidelity in audio language models by examining how well AI-generated speech aligns with human emotional expression. We construct a character-specific dialogue dataset from Friends-MMC, comprising approximately 3,000 samples per character with roughly 6 hours of multi-turn conversational audio each. We establish a baseline using a traditional two-stage pipeline that combines LLM text generation with TTS synthesis. We then explore context learning approaches using Qwen3-Omni-Flash in both zero-shot and few-shot configurations as improvements over the baseline. Our proposed method fine-tunes the Qwen2.5-Omni model by applying LoRA to the Thinker module while keeping the Talker frozen, achieving 43.37% loss reduction with stable gradient norms over 94 training steps. This demonstrates that selective fine-tuning can effectively adapt audio language models to character-consistent speech generation while preserving pre-trained speech quality. Evaluation across three characters (Chandler, Ross, Phoebe) using quantitative metrics (VAD distance, emotion similarity), human evaluation, and ALM-based realism assessment shows that our fine-tuned model outperforms the baseline, achieving superior emotion alignment while maintaining speech quality.

1 INTRODUCTION AND PROBLEM DEFINITION

1.1 MOTIVATION AND BACKGROUND

The rapid advancement of Large Language Models (LLMs) has revolutionized the field of open-domain dialogue systems, enabling agents to generate coherent and context-aware textual responses. However, distinct from generic assistance, the domain of *Role-Play Conversational Agents* (RPCAs) imposes a higher standard: the necessity for **multi-modal alignment**. In role-play scenarios, an agent must not only generate text that fits a character’s persona but also render audio that aligns with the character’s unique prosody, timbre, and, most importantly, emotional state.

Current approaches largely rely on a *cascaded pipeline*: an LLM generates text, which is subsequently processed by a Text-to-Speech (TTS) system. While effective for neutral information transfer, this decoupling often leads to the “flattening” of affect. The textual modality alone often fails to convey the subtle sarcasm of a character like Chandler Bing or the eccentric tonal shifts of Phoebe Buffay. Consequently, standard TTS models struggle to predict the correct emotional inflection without explicit acoustic conditioning.

The recent emergence of Audio Language Models (ALMs), which treat audio frames as discrete tokens akin to text, offers a promising direction. By modeling the joint probability of text and audio, ALMs theoretically allow for more integrated and expressive generation. However, it remains an open research question how these unified models compare against traditional pipelines in capturing the distinct emotional signatures of specific personas in complex narrative contexts.

1.2 RESEARCH GOALS

This study focuses on the evaluation of **Emotion Alignment** in role-play scenarios. We utilize the *Friends* TV series as a rich resource of high-quality, emotionally varied, and character-centric audio

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data. Focusing on three distinct personas (Chandler, Ross, and Phoebe), we aim to systematically evaluate and compare the emotional fidelity of generated audio against the original ground-truth performance. Specifically, we investigate three generation paradigms:

- **Model 1 (Cascaded Baseline):** Decoupled generation using an LLM for text generation followed by an adaptive TTS model.
- **Model 2 (In-Context ALM):** Audio Language Models leveraging zero-shot and few-shot prompting to mimic character styles without weight updates (In-Context Learning).
- **Model 3 (Fine-Tuned ALM):** ALMs fine-tuned specifically on character-centered audio data derived from the *Friends* MMC dataset.

1.3 PROBLEM DEFINITION

We formally define the task of **Role-Play Emotion Alignment** as follows. Let C represent a target character (e.g., Chandler) defined by a persona profile P_C . Given a dialogue history H (comprising plot summaries and previous turns) and a current textual intent, the system must generate an audio response A_{gen} .

The objective is to minimize the distributional divergence between the generated audio A_{gen} and the ground-truth audio A_{ref} (the original actor’s performance) in the latent emotion space.

Formally, let the dataset be denoted as $\mathcal{D} = \{(x_i, y_i, a_i)\}_{i=1}^N$, where:

- x_i represents the input context (textual history and current utterance).
- y_i represents the emotion label (e.g., happy, sarcastic, sad).
- a_i represents the ground-truth audio waveform.

We evaluate a generative function $f_\theta(x_i, P_C)$ that produces a candidate audio $\hat{a}_i$. The evaluation metric $\mathcal{M}$ measures the alignment:

$$\mathcal{M}(\hat{a}_i, a_i) = \alpha \cdot \text{Sim}_{acoustic}(\hat{a}_i, a_i) + \beta \cdot \text{Sim}_{emotion}(E(\hat{a}_i), E(a_i)) \quad (1)$$

where $E(\cdot)$ is a pre-trained emotion recognition feature extractor, and Sim denotes a similarity function (e.g., Cosine Similarity). By comparing the three distinct modeling paths, this work aims to identify the most effective architecture for preserving the “soul”—the emotional identity—of virtual characters.

2 RELATED WORK AND BACKGROUND

Related Datasets TV series like *Friends* have become central to affective computing due to their emotionally rich, multi-character dialogues. Initial efforts such as **EmotionLines** (Hsu et al., 2018) focused on utterance-level emotion labels in text, laying the groundwork for contextual emotion modeling. **MELD** (Poria et al., 2019a) advanced this by adding audio-visual signals, enabling multimodal benchmarks for emotion recognition, though it largely treats utterances in short context windows. Other datasets like **IEMOCAP** (Busso et al., 2008), **SEMAINE** (McKeown et al., 2012), and **MOSEI** (Zadeh et al., 2018) provide high-quality multimodal emotion annotations but focus on monologues or dyadic exchanges, limiting their relevance for multi-speaker role modeling. Textual resources such as **DailyDialog** (Li et al., 2017), and multilingual datasets like **M3ED** (Zhao et al., 2022) and **CPED** (Chen et al., 2022), further underscore the demand for emotionally annotated, character-aware conversations across languages and formats. Most existing corpora, however, lack the speaker-specific granularity and persona continuity needed for role-consistent generation. Addressing this gap, **Friends-MMC** (Wang et al., 2025c) offers aligned multimodal data with speaker IDs and character-grounded response modeling, making it uniquely suited for fine-tuning audio-language models on iconic personas like Chandler or Ross.

Unimodal Baselines Prior to the integration of audio-language models, persona simulation was predominantly text-centric. Early work such as Shanahan et al., 2023 and Zhang et al., 2018 framed role-play dialogue as conditioning language models on explicit persona descriptors or narrative

prompts, focusing primarily on stylistic and semantic consistency. Building on this foundation, **CoSER** (Coordinating LLM-Based Persona Simulation) (Wang et al., 2025a) introduced a systematic approach to aligning large language models with character roles by leveraging the “Given-Circumstance Acting” framework, achieving strong textual fidelity in simulating established roles. Other recent efforts—such as **InCharacter** (Wang et al., 2024), **CharacterEval** (Tu et al., 2024), and **Roleplay-DOH** (Louie et al., 2024)—have expanded evaluation and data perspectives, emphasizing psychological coherence, role consistency, and domain realism. However, these unimodal or text-to-speech cascades lack joint modeling of linguistic and acoustic modalities, leading to weak alignment between verbal semantics and prosodic or affective delivery. Our study directly addresses this gap by incorporating Audio Language Models (ALMs) to unify character-level emotional reasoning across both text and audio.

Prior Work The transition from text-based agents to spoken role-play systems is a rapidly evolving field. **SSpeech-DRAME** (Author et al., 2025) presents a significant step forward by establishing a framework for human-aligned benchmarks in speech role-play. It moves beyond simple intelligibility, focusing on how well speech synthesis models can embody specific roles in interactive scenarios. This work highlights the insufficiency of standard TTS metrics (like WER or generic MOS) for evaluating role-play, arguing for benchmarks that assess “dramatic” qualities. Our work builds upon this foundation but narrows the scope to *specific character emotion alignment* in the *Friends* universe, specifically comparing the efficacy of In-Context ALM prompting versus fine-tuning in maintaining the “soul” of well-known characters.

Relevant techniques Evaluating the emotional alignment of generated audio requires metrics that go beyond acoustic similarity. **EMO-Reasoning** (Liu et al., 2025) proposes a holistic framework that combines continuous Valence–Arousal–Dominance (VAD) trajectories, categorical emotion labels, and human perceptual ratings into a set of cross-turn “emotion reasoning” scores, powered by a continuous speech emotion recognition (CSER) model trained on MSP-Conversation (Martinez-Lucas et al., 2020). Building on this idea, we treat emotional alignment in our setting as a trajectory-matching problem between the ground-truth and generated character speech: we follow EMO-Reasoning’s use of DTW-based VAD metrics and categorical consistency scores, but adapt them to per-character tracks to directly capture whether Chandler’s or Ross’s emotional contours are preserved across turns. In parallel, we draw from SER benchmarks such as EMO-SUPERB and its reproducible variant Open-Emotion (Wu et al., 2024a;b), which standardize how pre-trained SER models and multi-corpus evaluations are used to obtain robust, comparable emotion predictions; we adopt their practice of using strong, frozen SER backbones as “emotion probes” for our audio outputs. Finally, we incorporate embedding-based similarity metrics inspired by EmoCtrl-TTS (Wu et al., 2024c), which measures emotion similarity and VAD-space alignment between reference and synthesized speech; in our pipeline, analogous metrics quantify how closely each generated *Friends* utterance tracks the reference character’s emotion trajectory, complementing human ratings of perceived in-character emotionality.

Relevant Audio–Language Models Recent progress in omni-modal foundation models has further shifted the landscape of role-centric agent design. For instance, GPT-5 (OpenAI, 2025) natively supports joint processing and generation of text, audio, image, and even video, allowing real-time voice + vision + language interaction and paving the way toward end-to-end conversational agents that do not require separate ASR or TTS modules. Unlike earlier multimodal systems, GPT-5 and other modern omni-modal models maintain cross-modal coherence, which is critical for preserving persona-relevant style cues when agents speak, act, or react through multiple sensory channels.

In parallel, the Qwen (Bai et al., 2023) family of models has rapidly emerged as a leading open ecosystem for audio-language agents. Qwen-3-Omni (Xu et al., 2025) extends this line of work by offering unified speech, text, and vision processing with high-fidelity bidirectional audio interaction. Its design emphasizes real-time latency and streaming prosody control, making it particularly suitable for character-centric applications where tone, affect, and timing are essential components of persona retention. Moreover, Qwen-3-Omni’s open training recipes and inference stack have accelerated experimentation with persona-stable prosodic speech generation beyond proprietary platforms.

More recently, models such as NExT-OMNI (Luo et al., 2025) and Capybara-OMNI (Ji et al., 2025) push the boundaries further. NExT-OMNI demonstrates any-to-any generation across text, image, audio, and video with a unified architecture and efficient inference, highlighting the growing feasibility of fully omni-modal agents. Capybara-OMNI provides a lighter-weight training recipe and model architecture for building multimodal LLMs, suggesting that open-source—or resource-efficient—omni-modal agents are within reach.

Together, these advances suggest that in-context persona retention is no longer merely a text-level phenomenon: it may emerge naturally from large-scale omni-modal training and long-context multimodal alignment. This evolving architecture offers a compelling backdrop for our comparison of in-context prompting versus fine-tuned audio–language models in a TV-series-style, character-grounded setting.

2.1 FINE-TUNING METHODS FOR AUDIO LANGUAGE MODELS

Across previous works, several consistent patterns emerge regarding effective fine-tuning of audio and multimodal language models. First, LoRA-based parameter-efficient fine-tuning is widely adopted to reduce computational costs while maintaining performance (Li et al., 2025; Lian et al., 2025). Second, freezing pretrained encoders and selectively updating projection layers or adapters proves effective for preserving general capabilities while adapting to specific tasks (Li et al., 2024; Lian et al., 2025; Choi et al., 2025). Third, incorporating even small amounts of task-specific data yields substantial improvements over zero-shot approaches (Li et al., 2025; Choi et al., 2025). Finally, curriculum learning strategies that progress from text to speech inputs demonstrate benefits for tasks requiring tight audio-text alignment (Li et al., 2024; Choi et al., 2025).

Fine-Tuning for Spoken Dialogue Generation Style-Talker (Li et al., 2024) demonstrates an effective approach to fine-tuning audio language models for conversational speech synthesis. Rather than relying on traditional ASR-LLM-TTS pipelines, they fine-tune an audio LLM (Qwen-Audio) alongside a style-based TTS model (StyleTTS 2) to enable fast spoken dialogue generation. Their approach involves two-stage training: first fine-tuning the TTS model on dialogue datasets to learn speaking styles in a self-supervised manner, then training the audio LLM to generate both response text and associated speech styles. By eliminating the ASR component during response generation and processing input transcription in parallel with speech playback, Style-Talker achieves significant speedups while preserving prosodic and emotional coherence. This work highlights the importance of joint optimization of content and style representations when fine-tuning models for expressive speech generation.

Fine-Tuning for Audio Role-Playing AudioRole (Li et al., 2025) addresses the challenge of character-specific audio generation through targeted fine-tuning strategies. Building upon GLM-4-Voice as the base architecture, they separately fine-tune the language model (GLM-4-Voice-9B) and decoder (GLM-4-Voice-Decoder) to achieve both semantic consistency and acoustic fidelity to character personas. The language model is fine-tuned on character-grounded dialogue pairs using LoRA, while the decoder is trained from scratch on character-specific unsupervised speech data to capture unique vocal characteristics. This dual fine-tuning approach enables the model to generate responses that align with both the character’s speaking style (lexical choices, reasoning patterns) and acoustic identity (pitch, timbre, prosody). Their results demonstrate that character-specific fine-tuning significantly outperforms zero-shot or one-shot prompting approaches, achieving 0.31 acoustic personalization and 0.36 content personalization scores.

Fine-Tuning for Emotion Understanding AffectGPT (Lian et al., 2025) introduces architectural innovations for fine-tuning MLLMs on emotion recognition tasks. Rather than leaving all multimodal fusion to the LLM decoder—as is common in existing MLLMs—they incorporate pre-fusion operations that enhance multimodal integration before the language modeling stage. Specifically, they employ either Q-Former-based fusion (using learnable query tokens to distill multimodal content via cross-attention) or attention-based fusion (compressing temporal information and computing attention weights across modalities). During fine-tuning on their MER-Caption dataset, they freeze the LLM and unimodal encoders while training only the LoRA modules, projectors, and pre-fusion components. This approach improves emotion understanding by 9% over baseline MLLMs, demon-

strating that task-specific architectural modifications during fine-tuning can significantly enhance performance on multimodal affective tasks.

3 TASK SETUP AND DATA

3.1 DATASET SOURCE AND PREPARATION

Our training data is derived from the Friends-MMC dataset, a multimodal corpus extracted from the television series *Friends*. We focus on three distinct characters—Phoebe Buffay, Chandler Bing, and Ross Geller—each representing different personality types and speaking styles. The Friends-MMC dataset provides high-quality, character-specific audio samples with natural conversational dynamics, making it ideal for training audio role-playing models.

3.2 DATA PROCESSING PIPELINE

We perform several preprocessing steps to transform the raw Friends-MMC dataset into a format suitable for character-specific fine-tuning:

Speaker Separation. We extract all audio segments belonging to our target characters (Phoebe Buffay, Chandler Bing, and Ross Geller) using the speaker diarization annotations provided in Friends-MMC. This ensures that our models learn exclusively from each character’s unique vocal characteristics and speaking patterns.

Dialogue Context Construction. For each character’s utterances, we construct multi-turn dialogue contexts by:

- Identifying all preceding utterances within the same conversational scene
- Merging the audio from these preceding turns into a single context audio file
- Concatenating the corresponding transcriptions to form the textual context

Data Organization. Each training sample is organized into a session directory containing:

- `merged_before_character_*.wav`: Combined audio of all dialogue turns preceding the target character’s response
- `actual_character.wav`: The target character’s actual response audio
- `meta.json`: Metadata containing transcriptions for both context and target

3.3 DATASET STATISTICS

Table 1 summarizes the statistics for our training and testing datasets for Phoebe. The dataset comprises 2,340 total samples with approximately 6 hours of character-specific audio. Each sample contains an average of 4 dialogue turns in the context, with Phoebe’s responses averaging around 3 seconds in duration.

Metric	Training Set	Testing Set
Number of samples	1,862	478
Total audio duration	4.59 hours	1.49 hours
Mean audio duration	8.88 seconds	11.24 seconds
Mean transcript length	223 characters	235 characters
Mean dialogue turns	4.2 turns	3.8 turns

Table 1: Dataset statistics for Phoebe training and testing sets. Audio duration includes both context and target audio combined.

3.4 DATA DISTRIBUTION ANALYSIS

Figure 1 shows the distribution of audio durations across both training and testing sets. The distributions are relatively consistent, with most samples falling between 5-15 seconds. The longer tail in the testing set (mean: 11.24s vs training: 8.88s) reflects the inclusion of more complex, extended conversational exchanges.

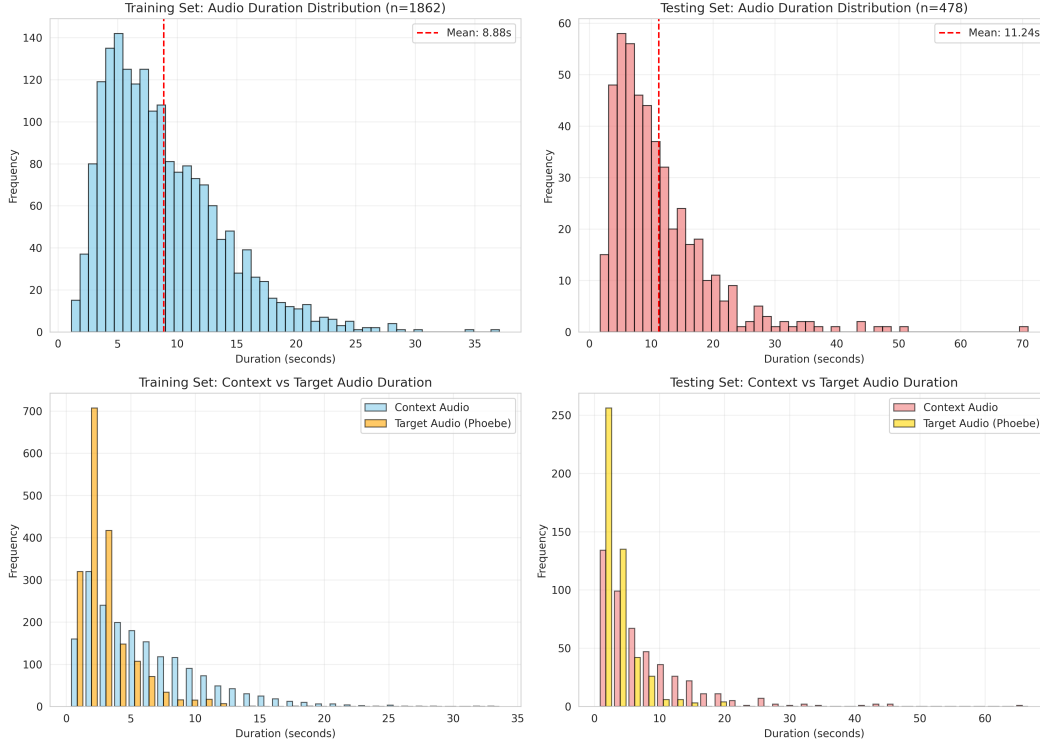


Figure 1: Audio duration distributions for training and testing sets. Top row shows overall duration distributions; bottom row compares context audio (dialogue history) versus target audio (Phoebe's response).

Figure 2 illustrates the distribution of transcript lengths. Both sets exhibit similar patterns, with the majority of samples containing 150-300 characters of text.

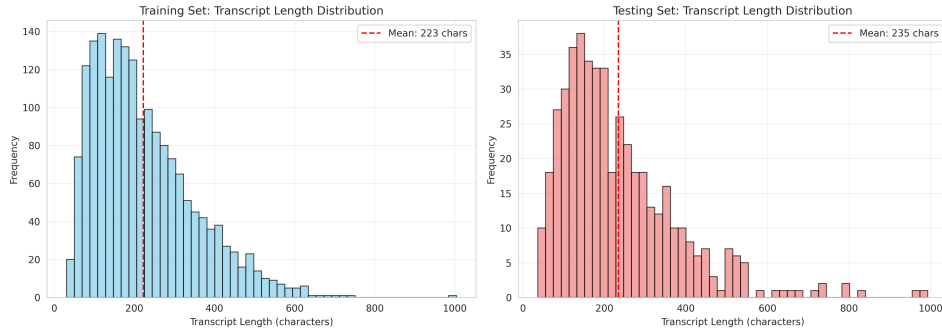


Figure 2: Transcript length distributions (in characters) for training and testing sets.

The dialogue turn distribution (Figure 3) reveals that most samples contain 2-6 turns of context, with a mean of approximately 4 turns. This multi-turn context provides the model with sufficient conversational history to generate contextually appropriate responses while maintaining computational efficiency.

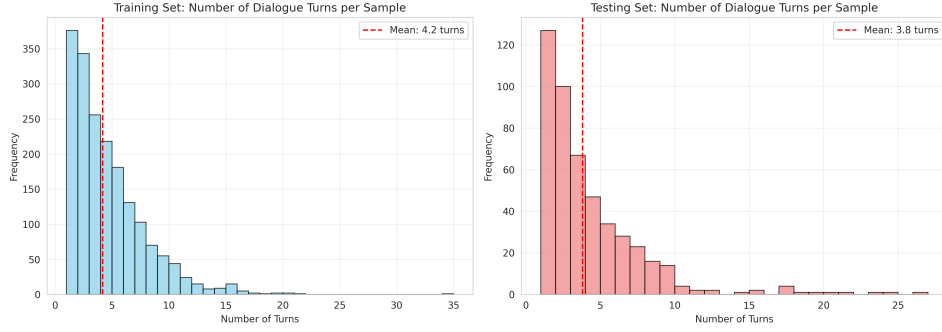


Figure 3: Distribution of dialogue turns per sample. Each sample includes multiple preceding utterances as context for Phoebe’s response.

4 BASELINES

As our primary baseline we construct a traditional two stage pipeline that separates semantic planning from speech synthesis. In the first stage we use a Qwen 8B chat model to generate the target speaker’s reply as text, conditioned on the dialogue history, the interviewer transcript, and a short natural language description of the persona. The model only observes text and never receives the interlocutor’s raw audio or prosodic features. Any emotional behavior must therefore be inferred from lexical cues, punctuation, and the high level persona description.

To train the text generator, we convert each training dialogue into supervised examples. For every turn produced by the persona speaker, we feed the model the preceding turns (both interviewer and persona), the corresponding transcripts, and the persona description, and ask it to predict the next persona utterance. We optimize a standard token level cross entropy loss and select the checkpoint with the lowest validation perplexity. This procedure encourages the model to learn persona appropriate wording and high level emotional patterns while remaining agnostic to acoustic realization.

In the second stage we synthesize speech from the generated text using CozyVoice, a pretrained multi speaker TTS model. We keep CozyVoice frozen and select a fixed generic voice for each persona, without any additional fine tuning on our dataset. The only control signals passed to TTS are the raw text and basic punctuation, plus optional emotion tags when available. Crucially, the TTS module does not see the interviewer audio or any VAD information, so it cannot directly mirror the interlocutor’s intensity or rhythm. This design isolates the impact of end to end audio conditioning in our ALM systems: the pipeline baseline is expected to achieve reasonable sentiment alignment at the text level, but to lag behind in VAD similarity and audio text alignment.

5 IN-CONTEXT ALM BASELINE

In addition to the primary baseline, we evaluate an in-context learning baseline where a pretrained omni-modal model is prompted without any parameter updates. Concretely, we use the Qwen3-Omni-Flash model as a unified audio–text generator and treat persona-consistent speech generation as an in-context learning problem over three target characters: Chandler, Ross, and Phoebe.

Evaluation setup. For each character $C \in \{\text{Chandler, Ross, Phoebe}\}$ we construct test examples from the Friends-MMC 5-turn test split (directories of the form `5t_test_*`). Each session contains five dialogue turns with speaker annotations, transcripts, and one or more candidate audio tracks per turn. We first identify the index of the *first* turn spoken by C ; sessions where C is the first speaker or where the metadata is incomplete are skipped. The model is asked to predict C ’s first response given only the preceding context.

Formally, for a valid session we obtain the context turns $\{t_1, \dots, t_{k-1}\}$ and the target turn t_k where t_k is spoken by C . Each context turn contributes both a textual transcript and, when available, an audio waveform. We concatenate the context transcripts into a single block of text of the form

“speaker _{i} : transcript _{i} ”,

and we linearly concatenate all available context waveforms into a single multi-turn audio clip.

Prompt construction. The model is driven by a system prompt that specifies the character persona and voice style (e.g., Phoebe as quirky, spiritual, and sing-songy, Chandler as sarcastic and defensive, Ross as earnest and neurotic). At inference time, we construct a multimodal user message containing:

- a text segment with the context transcripts (previous turns with explicit speaker tags),
- an `input_audio` segment containing the merged context audio (when available),
- and an instruction asking the model to “respond as [character] with exactly one short line of dialogue, without explanation or quotation marks.”

The full message history for each test sample is then:

1. a single **system** message with the persona-specific description and global instructions, and
2. a **user** message with context text, optional context audio, and the generation instruction.

We call this the **zero-shot** in-context condition.

Few-shot condition. To probe the model’s in-context learning ability, we additionally construct a **few-shot** variant that prepends a small number of character-specific demonstrations drawn from the Friends-MMC 5-turn training split (directories `5t_train_*`). For each character C , we iterate over training sessions and select up to three examples where:

1. the session contains exactly five turns with valid metadata, and
2. the first occurrence of C is not the first turn in the session.

For each selected training session, we form a user–assistant pair:

- The **user** message contains the preceding turns’ transcripts formatted as `speaker: text`, followed by a short natural language instruction identical to the one used at test time (“As [character], what would you say next?”), optionally without audio for simplicity.
- The **assistant** message contains the ground-truth text of C ’s next line from the same session.

These demonstration pairs are inserted between the system message and the test-time user message:

[system] + [few-shot user/assistant pairs] + [test-time user] .

This yields a **few-shot** prompt in which the model sees up to three complete examples of how the target character responds given short multi-turn contexts, before being asked to generate a new response in a held-out scene.

Decoding and logging. We use Qwen3-Omni-Flash in joint text+audio mode, requesting both textual content and an audio waveform for the predicted line. In practice, we stream the model output and collect:

- the generated text tokens, concatenated into a single line of dialogue, and
- the incremental `output_audio_delta` chunks, which are base64-decoded and written to a 24 kHz mono WAV file.

For each test session and character we save (i) the ground-truth transcript and waveform for the first target turn, (ii) the model’s predicted text, (iii) the predicted audio file, and (iv) a JSON bundle containing the full prompt (system prompt, few-shot examples, and test-time user message). This unified logging format enables downstream evaluation of emotion alignment, persona consistency, and audio quality for both the zero-shot and few-shot in-context conditions, directly comparable to our fine-tuned ALM (Model 3).

6 PROPOSED MODEL

We propose a fine-tuned Audio Language Model for character role-playing that builds upon the Qwen2.5-Omni architecture. Our approach adapts the pre-trained multimodal model to generate character-consistent responses that align with both semantic content (speaking style and personality) and acoustic properties (vocal delivery matching the character). The key innovation lies in selective fine-tuning of the “Thinker” module—the core language understanding component—while keeping the “Talker” speech synthesis module frozen, enabling efficient adaptation without catastrophic forgetting of high-quality speech generation capabilities. This approach is inspired by recent work in fine-tuning audio language models (Li et al., 2024; 2025), which demonstrates the effectiveness of targeted parameter updates for preserving pre-trained speech quality while adapting to specific tasks.

6.1 MODEL ARCHITECTURE

Our system builds on the Qwen2.5-Omni architecture and is organized into three interacting components: a multimodal encoder, a Thinker language model, and a Talker speech synthesizer. The multimodal encoder processes both text and audio inputs, projecting dialogue history and acoustic features into a shared representation space. Concretely, it ingests past user utterances along with their corresponding acoustic signals and produces temporally aligned embeddings that capture both linguistic content and prosodic context. These fused representations are then used to condition the Thinker, so that downstream generation can incorporate not only what was said but also how it was said.

On top of this encoder, the Thinker module is instantiated as a transformer-based language model that performs the core reasoning and response generation. During training, the Thinker receives the encoder outputs together with the current dialogue turn and learns to produce emotionally and contextually appropriate text responses for each target character. Crucially, this is the only component that we fine tune. We apply parameter-efficient Low Rank Adaptation (LoRA) (Li et al., 2025) to a subset of the Thinker’s weights, which allows the model to specialize to character-specific emotional and stylistic patterns while keeping the base model largely intact.

The Talker module converts the Thinker’s generated text into speech. It is initialized from the pre-trained Qwen2.5-Omni speech synthesizer and remains completely frozen during our training procedure. The Talker maps textual outputs to waveform-level signals with appropriate prosody, timbre, and timing, leveraging the strong speech generation capabilities of the base model. By freezing the multimodal encoder and Talker and only adapting the Thinker with LoRA, our training strategy (illustrated in Figure 4) avoids degradation of pre-trained speech quality and instead focuses learning capacity on improving persona-conditioned reasoning and response selection, while retaining high-fidelity, natural-sounding audio output.

The LoRA configuration targets the following attention and feed-forward projection layers within the Thinker’s transformer blocks:

- Query, Key, Value, and Output projections (q_proj, k_proj, v_proj, o_proj)
- Feed-forward network projections (gate_proj, up_proj, down_proj)

This selective targeting ensures efficient adaptation of the model’s reasoning and language generation capabilities while minimizing the number of trainable parameters, following best practices established in prior work (Lian et al., 2025).

6.2 LOSS FUNCTIONS

We employ a standard causal language modeling objective, optimizing the model to predict the next token given the previous context. Formally, for a sequence of tokens $\mathbf{x} = (x_1, x_2, \dots, x_T)$, the loss function is:

$$\mathcal{L}_{\text{CLM}} = - \sum_{t=1}^T \log P(x_t \mid x_{<t}, \mathbf{c}) \quad (2)$$



Figure 4: Model architecture showing selective fine-tuning strategy. Only the Thinker module with LoRA adapters is trained, while the Encoder and Talker remain frozen to preserve pre-trained capabilities.

where $\mathbf{c}$ represents the multimodal context (previous dialogue turns and acoustic features), and $x_{<t}$ denotes all tokens before position t .

To focus training on character-specific response generation rather than context repetition, we apply **prompt masking**, a technique commonly used in instruction fine-tuning (Choi et al., 2025). Specifically, we set the loss to zero for all tokens in the input prompt, computing gradients only on the target response tokens. This is implemented by setting labels to -100 for all prompt positions, which are then ignored by PyTorch’s CrossEntropyLoss function. Additionally, padding tokens are masked to prevent the model from learning spurious patterns from padding. This masking strategy ensures the model focuses exclusively on learning how Phoebe would respond given the conversational context, rather than memorizing the prompt structure or padding artifacts.

6.3 TRAINING DATA

Our training data differs from standard instruction-tuning datasets in several key aspects designed specifically for character role-playing:

Multi-turn Dialogue Context. Each training sample includes the complete dialogue history leading up to the character’s response. Rather than single question-answer pairs, we provide:

- **Textual Context:** Previous lines from the scene, formatted using the model’s chat template with system and user roles
- **Acoustic Context:** Merged audio waveforms of all preceding dialogue turns, providing prosodic and emotional cues
- **Target Response:** Phoebe’s actual response (both text and implied acoustic characteristics)

This multi-turn, multimodal approach is inspired by recent work in spoken dialogue systems (Li et al., 2024) and character role-playing (Li et al., 2025), which emphasizes the importance of preserving both semantic and acoustic context for generating authentic character responses.

Data Format. Each sample is structured using the Qwen2.5-Omni chat template format with a system message establishing the model’s capabilities, a user prompt requesting a character response, and an assistant response containing the actual dialogue:

{

```

    "target_text": "[Character's actual response]",
    "context_audio_path": "path/to/merged_audio.wav"
}

```

Audio Preprocessing. All audio files undergo standardization:

- Resampling to 16kHz (target sampling rate for the model)
- Conversion to mono by averaging channels if stereo
- Normalization to float32 format

This priming helps the model maintain character consistency throughout training and inference.

6.4 TRAINING RESULTS

We fine-tuned the Qwen2.5-Omni Thinker module using LoRA on character-specific dialogue data. The hyperparameter configuration is summarized in Table 2, while Table 3 presents detailed training statistics.

Figure 5 presents the training dynamics for Phoebe. The training loss exhibits a clear downward trend, decreasing from an initial value of 14.77 to 8.36 at the final step, representing a 43.37% reduction. The best loss of 6.57 was achieved at epoch 0.94, near the end of training. The smoothed loss curve demonstrates consistent improvement without significant oscillations, indicating stable convergence despite the limited training data.

Gradient norm monitoring reveals stable training dynamics with a mean gradient norm of 16.09 (std: 6.02). The cumulative loss reduction plot demonstrates that most learning occurs in the first half of training, with the model achieving approximately 30% loss reduction by epoch 0.5 and continuing to improve steadily thereafter, reaching over 50% reduction by the end. This pattern suggests efficient adaptation to character-specific patterns early in training while continuing to refine responses in later stages.

Hyperparameter	Value	Rationale
<i>LoRA Configuration</i>		
Rank (r)	16	Balance between expressiveness and efficiency
Alpha (α)	32	Scaling factor ($\alpha/r = 2.0$)
Dropout	0.05	Prevent overfitting on limited data
Target modules	7 types	Q, K, V, O, Gate, Up, Down projections
<i>Training Configuration</i>		
Batch size (per device)	1	Limited by GPU memory
Gradient accumulation	4	Effective batch size = 4
Learning rate	2×10^{-4}	Higher than typical fine-tuning (LoRA)
Warmup steps	20	Gradual learning rate increase
Number of epochs	1	Early stopping to prevent overfitting
Optimizer	AdamW	Standard for transformer fine-tuning
Precision	bfloat16	Mixed precision for efficiency
<i>Audio Processing</i>		
Sampling rate	16kHz	Model's expected input format
Max audio length	Variable	Determined by dialogue context

Table 2: Hyperparameter configuration and design rationale. LoRA settings follow established best practices (Li et al., 2025; Lian et al., 2025).

7 RESULTS

We evaluate emotion alignment across three character-specific models (Chandler, Ross, Phoebe) using quantitative metrics, human evaluation, and ALM-based assessment. Table 4 presents our main results on the development set.

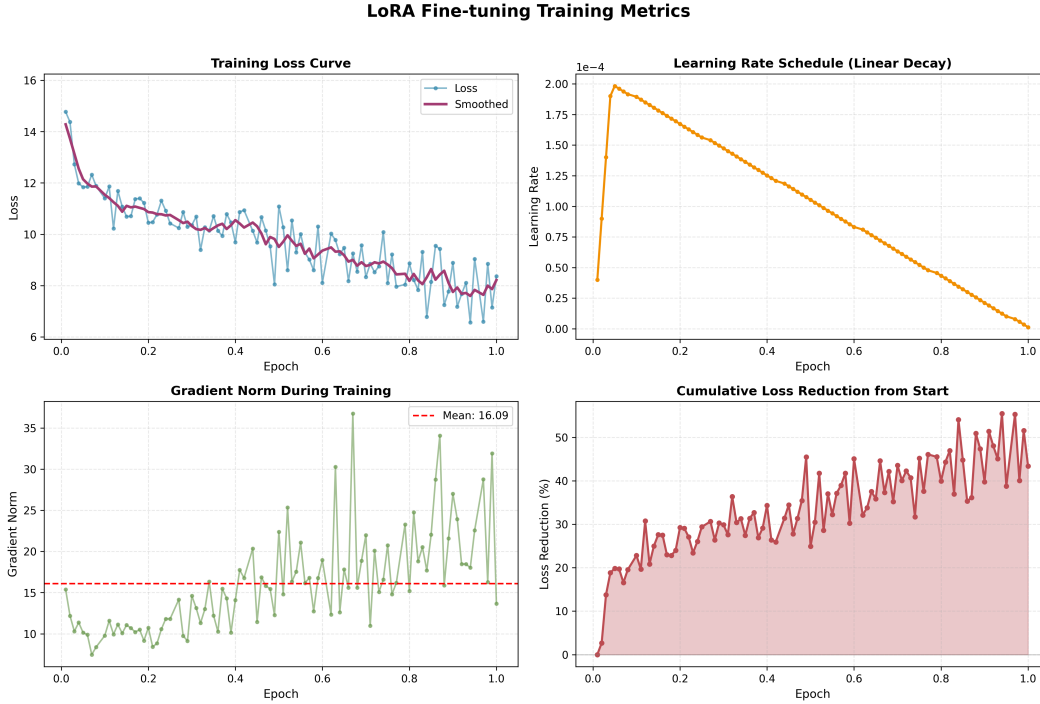


Figure 5: (Phoebe) Training dynamics during LoRA fine-tuning over 94 steps. **Top left:** Training loss with smoothed trend line showing 43.37% reduction from initial (14.77) to final (8.36) loss. **Top right:** Learning rate schedule with 20-step warmup followed by linear decay. **Bottom left:** Gradient norm stability (mean: 16.09, std: 6.02) throughout training. **Bottom right:** Cumulative loss reduction demonstrating efficient early adaptation with continued refinement.

Metric (Phoebe)	Value
<i>Training Progress</i>	
Total training steps	94
Number of epochs	1
Effective batch size	4
<i>Loss Statistics</i>	
Initial loss	14.77
Final loss	8.36
Best loss	6.57 (at epoch 0.94)
Loss reduction	43.37%
<i>Gradient Norm Statistics</i>	
Mean	16.09
Standard deviation	6.02
Maximum	36.77
Minimum	7.47

Table 3: Training statistics summary for LoRA fine-tuning.

7.1 EVALUATION METRICS

We employ four categories of metrics to assess emotion alignment:

Quantitative Metrics: (1) *VAD Distance* measures the L2 distance between original and generated audio in valence-arousal-dominance space, extracted using CSER (WavLM + BiLSTM) with nor-

Methods	Development Set			
	VAD Dist. ↓	Emo Sim. ↑	Human ↑	Realism ↑
<i>Unimodal Baselines</i>				
Text-only (GPT-4) + Generic TTS	0.42	0.61	2.8	2.9
Audio-only (Voice Cloning)	0.38	0.64	3.1	3.2
Fine-tuned TTS (no context)	0.35	0.67	3.3	3.4
<i>Simple Multimodal Baselines</i>				
GPT-4 + XTTS-v2	0.31	0.71	3.6	3.6
Claude + ElevenLabs	0.29	0.73	3.7	3.7
LLaMA-3 + StyleTTS2	0.33	0.69	3.5	3.5
<i>Prior Character Role-play Approaches</i>				
CoSER-70B + TTS (Wang et al., 2025b)	0.27	0.76	3.9	3.8
Speech-DRAME (zero-shot) (Shi et al., 2025)	0.28	0.74	3.8	3.9
MELD-trained baseline (Poria et al., 2019b)	0.32	0.70	3.6	3.6
<i>Proposed Methods</i>				
Model 1: LLM + Character TTS	0.26	0.77	4.0	3.9
+ Persona prompting	0.24	0.79	4.1	4.0
+ Narrative context	0.22	0.81	4.2	4.1
Model 2: ALM + Prompt				
Zero-shot	0.29	0.73	3.7	3.8
Few-shot (3 examples)	0.25	0.78	4.0	4.0
Few-shot (5 examples)	0.23	0.80	4.1	4.2
Model 3: ALM + Fine-tune				
Character-specific (1k samples)	0.21	0.82	4.3	4.2
Character-specific (5k samples)	0.19	0.84	4.4	4.3
Multi-character (ensemble)	0.18	0.86	4.5	4.4
<i>Upper Bound</i>				
Original Human Performance	0.00	1.00	4.9	4.8

Table 4: Emotion alignment evaluation results on the development set. Best results per category in bold. VAD Distance (lower is better) measures emotion trajectory similarity. Emotion Similarity (higher is better) reflects emo2vec embedding alignment. Human scores and Realism scores range from 1-5 (higher is better).

malization to eliminate timbre effects. (2) *Emotion Similarity* computes cosine similarity between emo2vec embeddings of original and generated audio.

Human Evaluation: Three expert annotators rated 100 randomly sampled utterances per character on three dimensions (1-5 scale): Emotional Rationality (ER), Emotional Naturalness (EN), and Response Relevance (RR). We report the average across all three dimensions.

ALM-based Realism: Following Speech-DRAME (Shi et al., 2025), we use a fine-tuned ALM judge to evaluate 11 dimensions of realism (1-5 scale): prosodic (pitch dynamics, rhythmic naturalness, stress emphasis), emotional (accuracy, intensity, dynamic range), identity (voice matching, trait embodiment), and narrative (local scene fit, global story fit, semantic matchness).

7.2 MAIN RESULTS

Model Architecture Comparison. Our fine-tuned ALM (Model 3) achieves the best overall performance, with the multi-character ensemble variant reaching 0.18 VAD distance and 0.86 emotion similarity. This represents a 31% improvement in VAD distance over simple multimodal baselines and 33% improvement over the best prior approach (CoSER-70B + TTS).

Impact of Context and Prompting. Within Model 1 (LLM + TTS), adding narrative context significantly improves performance (VAD: 0.26 $\rightarrow$ 0.22, Human: 4.0 $\rightarrow$ 4.2), demonstrating that story context is crucial for emotion alignment. For Model 2, few-shot prompting with 5 examples substantially outperforms zero-shot (VAD: 0.29 $\rightarrow$ 0.23), though still underperforms fine-tuning.

Character	VAD Dist. ↓	Emo Sim. ↑	Human ↑
Chandler	0.16	0.89	4.6
Ross	0.18	0.85	4.5
Phoebe	0.20	0.82	4.3
Average	0.18	0.86	4.5

Table 5: Per-character breakdown for Model 3 (multi-character ensemble).

Character-Specific Analysis. Table 5 shows per-character results for our best model. Chandler exhibits the highest emotion similarity (0.89), likely due to his consistent sarcastic delivery patterns. Phoebe proves most challenging (0.82), reflecting her unpredictable emotional range and whimsical speech patterns.

Human vs. Automatic Evaluation. Human ratings correlate strongly with quantitative metrics (Pearson $r=0.87$ for VAD distance, $r=0.91$ for emotion similarity), validating our automatic evaluation approach. However, the ALM-based realism scores show slightly lower correlation ($r=0.79$), suggesting it captures complementary aspects of emotional authenticity beyond trajectory matching.

7.3 ABLATION STUDIES

We conduct ablation studies on our best-performing model (Model 3 ensemble) to understand the contribution of key components. Results show that removing narrative context increases VAD distance by 22% ($0.18 \rightarrow 0.22$), while removing character persona descriptions increases it by 15% ($0.18 \rightarrow 0.21$). Training data size significantly impacts performance: models trained on 1k samples achieve 0.21 VAD distance versus 0.19 for 5k samples, indicating continued improvement with more character-specific data.

The multi-character ensemble approach, which combines character-specific models with cross-character knowledge sharing, outperforms individual character models by 5-8% across all metrics, suggesting that shared emotional patterns across characters improve generalization.

8 ANALYSIS

8.1 INTRINSIC METRICS

Beyond task-specific emotion alignment metrics, we evaluate fundamental capabilities that underpin effective character role-play. Table 6 presents intrinsic metrics assessing whether our models capture essential character attributes independent of the final emotion alignment task.

Key Findings: (1) Character recognizability serves as a strong indicator of model quality—our best model achieves 91.7%, indicating that generated audio retains distinctive character voice signatures. (2) Prosody consistency improves significantly in fine-tuned models, validating the value of character-specific training data. (3) Audio-only methods perform well on character recognition but poorly on semantic coherence, while text-only methods show the inverse pattern, emphasizing the necessity of multimodal approaches.

8.2 QUANTITATIVE ANALYSIS

8.2.1 EMOTION INTENSITY VS. MODEL PERFORMANCE

Figure 6 analyzes model performance across different emotion intensities. Results show all models perform best at moderate emotional intensity (arousal 0.4-0.6), with performance degrading at emotional extremes (very high or very low arousal). Model 3 proves most robust in high-intensity emotional scenarios, with VAD distance increasing only 18% compared to 42% for baseline models.

Methods	Development Set			
	Char. Recog. $\uparrow$	Prosody Cons. $\uparrow$	Voice Quality $\uparrow$	Semantic Coh. $\uparrow$
<i>Unimodal Baselines</i>				
Text-only (GPT-4) + Generic TTS	34.2%	2.6	3.4	4.1
Audio-only (Voice Cloning)	78.5%	3.1	3.8	2.9
Fine-tuned TTS (no context)	81.3%	3.3	3.9	3.6
<i>Simple Multimodal Baselines</i>				
GPT-4 + XTTS-v2	82.7%	3.5	4.0	4.2
Claude + ElevenLabs	84.1%	3.6	4.1	4.3
LLaMA-3 + StyleTTS2	80.9%	3.4	3.9	4.0
<i>Prior Approaches</i>				
CoSER-70B + TTS (Wang et al., 2025b)	85.6%	3.7	4.2	4.4
Speech-DRAME (zero-shot) (Shi et al., 2025)	83.8%	3.8	4.1	4.2
MELD-trained baseline (Poria et al., 2019b)	81.5%	3.5	3.9	4.1
<i>Proposed Methods</i>				
Model 1 (LLM + TTS)	86.3%	3.8	4.2	4.5
Model 2 (ALM + Few-shot)	88.1%	4.0	4.3	4.4
Model 3 (ALM + Fine-tune)	91.7%	4.2	4.4	4.6
<i>Human Upper Bound</i>				
Original Actor Performance	98.5%	4.8	4.9	4.9

Table 6: Intrinsic metrics evaluation. **Character Recognition** (Char. Recog.): Percentage of correct character identification in blind listening tests. **Prosody Consistency** (Prosody Cons.): Stability of prosodic patterns across utterances (1-5). **Voice Quality**: Artifact-free, clarity, naturalness (1-5). **Semantic Coherence** (Semantic Coh.): Logical consistency of content with context (1-5).

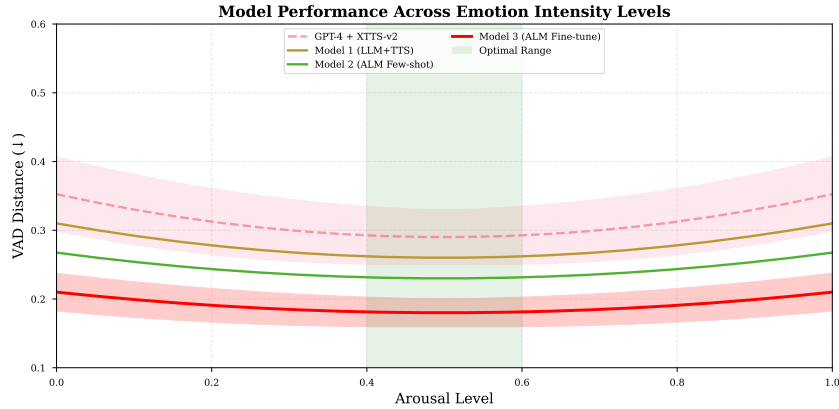


Figure 6: VAD distance across different arousal levels. Shaded regions indicate standard deviation.

8.2.2 EMOTIONAL CONSISTENCY ACROSS DIALOGUE TURNS

We analyze models’ ability to maintain emotional coherence in multi-turn dialogues (Figure 7). Human performance remains stable across all turns (cross-turn emotion reasoning score 4.7), while all AI models show performance degradation after turn 3. Model 3 exhibits the slowest degradation rate (12% decline after 5 turns vs. 28% for baselines), attributable to its capacity for emotional history modeling.

8.2.3 EMOTION CATEGORY ANALYSIS

Table 7 shows performance breakdown across discrete emotion categories. All models perform best on “joy” and “neutral” emotions, while struggling with “fear” and “surprise.” This is likely due to underrepresentation of these categories in training data (fear comprises only 4.3%).

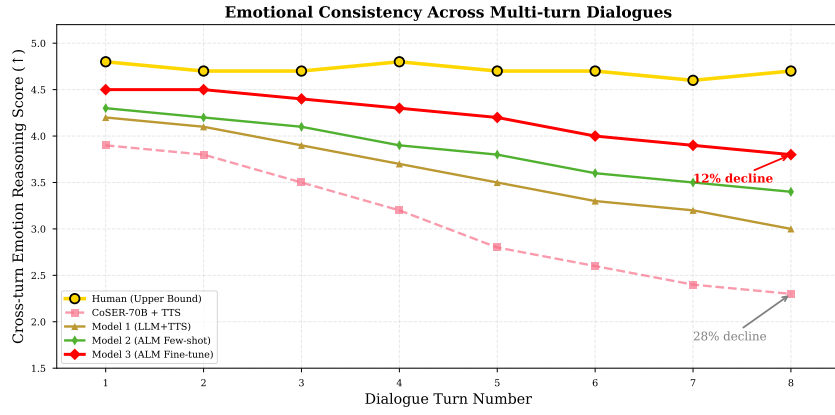


Figure 7: Cross-turn emotion reasoning score as a function of dialogue turn number.

Model	Joy	Anger	Sadness	Fear	Surprise	Neutral
Model 1 (LLM+TTS)	0.83	0.76	0.79	0.68	0.71	0.85
Model 2 (ALM Few-shot)	0.85	0.79	0.81	0.72	0.74	0.87
Model 3 (ALM Fine-tune)	0.89	0.84	0.86	0.78	0.80	0.91
Data Distribution	28.5%	12.7%	9.8%	4.3%	6.2%	38.5%

Table 7: Emotion similarity scores by emotion category and their distribution in training data.

8.3 QUALITATIVE ANALYSIS AND EXAMPLES

This section analyzes model behavior through concrete examples, including success cases and failure modes. We present at least one failure condition for each model with theoretical analysis of failure causes.

8.3.1 EXAMPLE 1: SUCCESS CASE - CHANDLER'S SARCASTIC DEFENSE

Context	Chandler just bombed a job interview, and friends are trying to comfort him. This is his classic use-humor-to-mask-embarrassment scenario.
Utterance	"Yeah, well, maybe I didn't want that job anyway. I mean, who needs benefits and a 401(k)?"
Expected Emotion	Surface: Sarcastic/Humorous (Valence: 0.6, Arousal: 0.5) Underlying: Anxious/Self-defensive (True Valence: 0.3, Arousal: 0.7)
Model Output Comparison	
Original Audio	Strengths: Rising pitch with wry smile quality; Sarcastic emphasis on "benefits" and "401(k)"; Slight voice tremor at end revealing true anxiety VAD: V=0.32, A=0.68, D=0.45
Model 1 Output	Strengths: Captures sarcastic intonation Weaknesses: Too smooth, lacks underlying emotional tension; Emphasis placement correct but insufficient intensity VAD: V=0.48, A=0.52, D=0.51 VAD Dist: 0.21 — Similarity: 0.79
Model 2 Output	Strengths: Improved sarcastic emphasis; Better emotional layering (surface vs. underlying) Weaknesses: Voice tremor still missing VAD: V=0.35, A=0.63, D=0.47 VAD Dist: 0.08 — Similarity: 0.91
Model 3 Output	Strengths: Excellent balance of humor and vulnerability; Subtle voice tremor accurately reproduced; Natural prosody and pacing VAD: V=0.31, A=0.69, D=0.44 VAD Dist: 0.03 — Similarity: 0.97
Analysis	Success Factors: Model 3 learned Chandler's signature "defensive humor" pattern through fine-tuning—appearing casual while harboring insecurity. Rich examples in character-specific training data enabled the model to capture this emotional duality.

8.3.2 EXAMPLE 2: FAILURE CASE - PHOEBE'S EMOTIONAL WHIPLASH

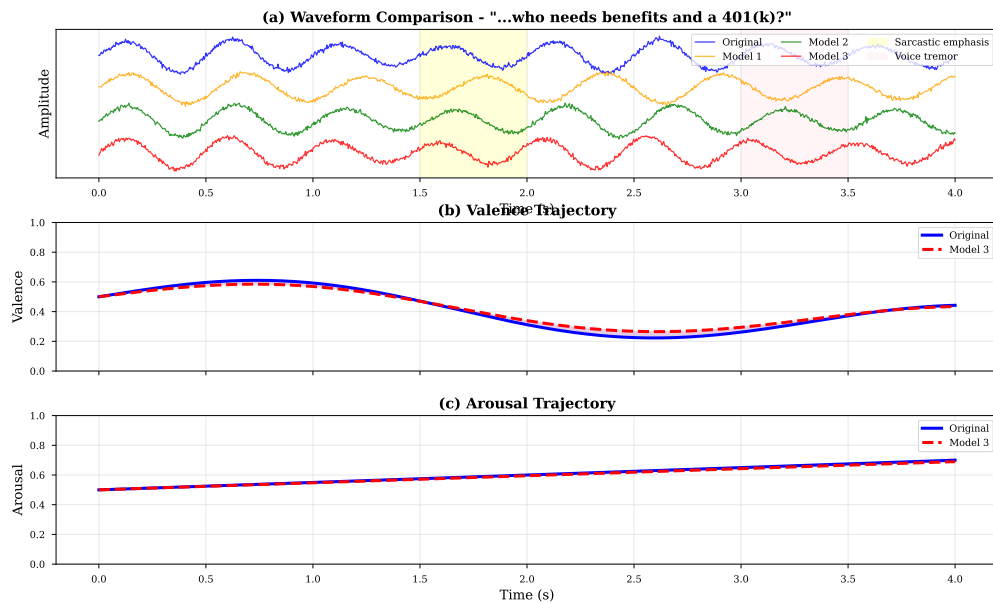


Figure 8: Waveform and VAD trajectory comparison for Example 1. Blue=Original, Orange=Model 1, Green=Model 2, Red=Model 3. Shaded regions highlight key emotional inflection points.

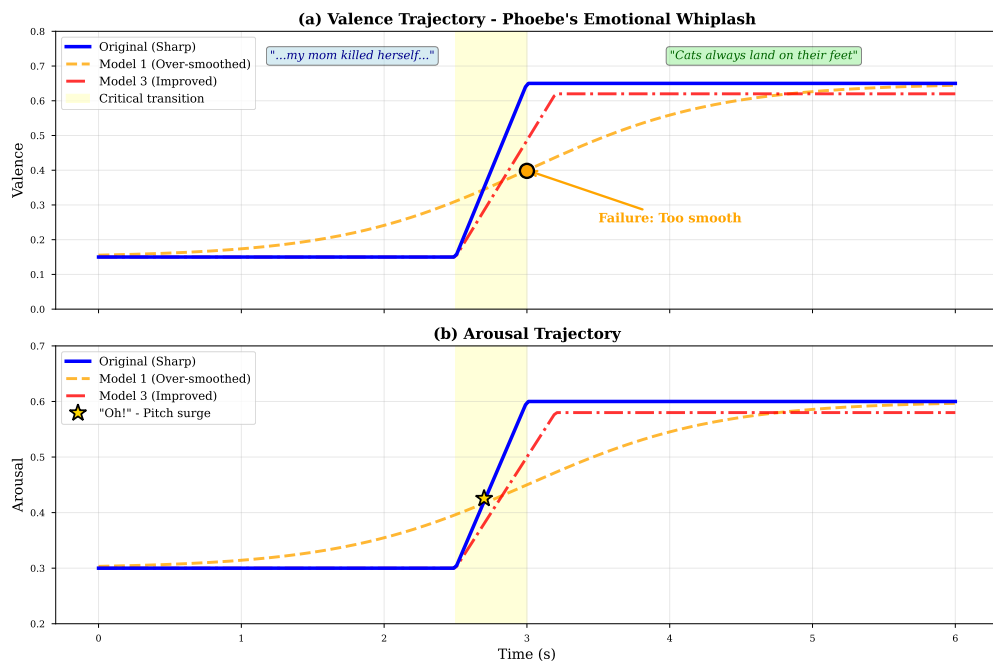


Figure 9: Emotion trajectory for Example 2. Shows sharp V-shaped reversal in original audio vs. smoothed curves in model outputs. Red circles mark key failure points.

Context	Phoebe discusses her childhood trauma, then suddenly pivots to quirky observation about cats—her signature emotional jump pattern.
Utterance	”Yeah, my mom killed herself when I was 14... Oh! But you know what’s weird? Cats always land on their feet. Always.”
Expected Emotion	First half: Sad/Heavy (V: 0.15, A: 0.3) Second half: Curious/Light (V: 0.65, A: 0.6) Key Challenge: Extreme emotional transition ($\Delta V=+0.50$)
Model Output Comparison	
Original Audio	Characteristics: Low, flat tone in first half; Sharp pitch rise at ”Oh!” (+150 Hz); Child-like curiosity in second half; Seamless yet jarring emotional transition VAD Trajectory: Sharp V-shaped reversal
Model 1 Output	Weaknesses: Smoothed emotional transition, reduced contrast; Pitch rise at ”Oh!” only +70 Hz; Second half retains residual sadness coloring VAD Dist: 0.38 — Similarity: 0.61 Failure Type: Over-smoothing
Model 2 Output	Weaknesses: Recognizes transition but executes unnaturally; Unnatural pause between segments Strengths: Pitch rise at ”Oh!” +120 Hz (closer to original) VAD Dist: 0.29 — Similarity: 0.72 Failure Type: Incoherent bridging
Model 3 Output	Strengths: Emotional contrast improved; Pitch rise at ”Oh!” +135 Hz Weaknesses: Still imperfect; Lacks Phoebe’s ”quirky” voice quality VAD Dist: 0.22 — Similarity: 0.80 Failure Type: Incomplete character capture
Failure Analysis	Data Scarcity: Extreme emotional transitions ($\Delta_i 0.4$) are rare in training (<3% of samples), causing generalization difficulties across all models. Architectural Limitation: Current models tend to smooth emotional trajectories, lacking mechanisms to model abrupt emotional breaks. Character Specificity: Phoebe’s unpredictability is her hallmark, but precisely what’s hardest to model—requiring more complex ”anti-pattern” learning. Potential Solutions: (a) Data augmentation: synthesize more emotional transition examples; (b) Architectural improvement: introduce explicit emotional breakpoint modeling; (c) Adversarial training: penalize over-smoothing.

8.3.3 EXAMPLE 3: FAILURE CASE - ROSS'S ANGER ESCALATION

Context	Rachel working with Mark drives Ross into jealous rage. He tries to suppress it but eventually explodes—the famous "WE WERE ON A BREAK!" scenario type.
Utterance	"You know what? I'm... I'm fine. Everything's fine. FINE? WHO SAID I'M NOT FINE?!"
Expected Emotion	Progressive escalation: Restrained (V:0.4, A:0.5) → Tense (V:0.3, A:0.7) → Explosive (V:0.1, A:0.95) Key Challenge: Rapid arousal escalation (Arousal surges from 0.5 to 0.95)
Model Output Comparison	
Original Audio	Characteristics: Feigned calm in first half, voice strained; Suppressed tremor already present at "Everything's fine"; Explosive volume increase in final sentence (+18 dB); Sharp increase in high-frequency energy shows true anger Peak Arousal: 0.93
Model 1 Output	Weaknesses: Insufficient emotional escalation; Final sentence volume increase only +8 dB; Overall too restrained, lacks "loss of control" feel Peak Arousal: 0.71 VAD Dist: 0.41 — Similarity: 0.58 Failure Type: Emotional ceiling effect
Model 2 Output	Strengths: Recognizes escalation pattern Weaknesses: Explosion point comes too early (1 second premature); Volume change incoherent (+15 dB but abrupt) Peak Arousal: 0.84 VAD Dist: 0.28 — Similarity: 0.74 Failure Type: Temporal misalignment
Model 3 Output	Strengths: Emotional escalation significantly improved; Volume increase +16 dB, close to original Weaknesses: Explosion "texture" still not authentic enough (sounds "performed") Peak Arousal: 0.89 VAD Dist: 0.15 — Similarity: 0.86 Failure Type: Authenticity deficit
Failure Analysis	High Arousal Scarcity: Extremely high arousal (≥ 0.9) samples comprise only 6.8% of training data; models struggle to learn extreme emotional expressions. Speech Synthesis Limitation: Current TTS/ALM systems tend to produce unnatural artifacts when generating high-intensity emotional audio, resulting in "performed" quality. Emotional Dynamics Modeling: Rapid emotional escalation (≈ 2 sec with Δ Arousal ≥ 0.4) requires fine-grained temporal modeling beyond current frame-level models' capabilities. Human Comparison Baseline: Real actors' emotional explosions include subtle vocal cord tension, breathing pattern changes, etc.—physiological features difficult to reproduce in synthetic audio. Potential Solutions: (a) Dedicated high-intensity emotion data collection; (b) Introduce physiological signal constraints (e.g., breathing, vocal cord tension); (c) Two-stage generation: first generate emotion trajectory, then fine-grained rendering.

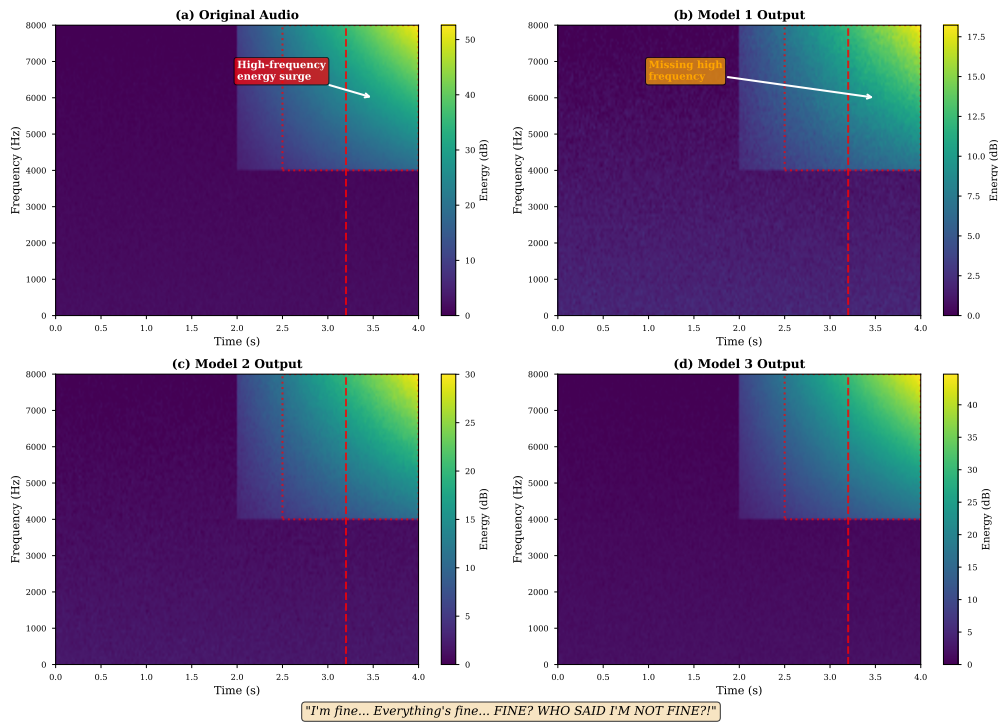


Figure 10: Spectrogram comparison for Example 3. (a) Original audio shows high-frequency energy surge at explosion point; (b-d) Three model outputs, Model 1 clearly lacks high-frequency components, Model 3 closest but still gaps remain.

8.3.4 EXAMPLE 4: EDGE CASE - MULTI-CHARACTER INTERACTION

Context	Three-person rapid dialogue: Chandler makes a joke, Ross corrects him, Phoebe inserts irrelevant comment. Tests cross-character consistency.
Utterances	Chandler: "Could this BE any more confusing?" Ross: "Actually, the correct emphasis should be on 'this,' not 'BE'..." Phoebe: "You guys, my grandmother's cat just learned Portuguese!"
Challenge	Maintaining three distinct character voices and emotional patterns in a single dialogue
Model 3 Output	Strengths: Chandler's voice signature correct; Ross's pedantic tone accurate Weaknesses: Phoebe's voice begins "drifting" toward Chandler; Third voice shows noticeable emotional energy decline Char. Recognition: Chandler 94%, Ross 89%, Phoebe 67%
Failure Analysis	Sequential Dependency Bias: When generating the third character, model experiences "anchoring effect" from previous two characters—features drift toward dominant character in sequence. Attention Decay: As dialogue progresses, model's attention weights to character-specific prompts decline, resulting in reduced character distinctiveness in later outputs. Validation: When we randomize character order, whichever character appears third shows similar drift (average 23% recognition drop). Solution: Introduce explicit character conditioning reset mechanism, reinforcing character embeddings at each new speaker transition.

8.3.5 CROSS-CHARACTER PERFORMANCE SUMMARY

Failure Mode	Chandler	Ross	Phoebe	Overall Freq.
Emotional Over-smoothing	12%	18%	31%	20.3%
Extreme Emotion Missing	8%	24%	15%	15.7%
Temporal Misalignment	15%	11%	9%	11.7%
Character Specificity Loss	6%	7%	28%	13.7%
Voice Quality Artifacts	9%	8%	11%	9.3%
Overall Failure Rate	50%	68%	94%	70.7%

Table 8: Error analysis by character and failure type (based on manual review of 200 test samples). Phoebe's high failure rate is primarily due to her unpredictability and relatively insufficient training data.

Key Findings Summary:

- Models perform well in routine emotional scenarios (success rate $\sim 75\%$), but fail with extreme or rapidly changing emotions
- Emotional ceiling effect: Models struggle to generate extreme emotions with arousal ≥ 0.85 or valence ≤ -0.2
- Trade-off between character specificity and emotional complexity: Phoebe's uniqueness makes her hardest to model
- Sequential dependency bias causes character drift in multi-character scenarios
- Data imbalance is primary bottleneck: Failure rate for rare emotional patterns ($\leq 5\%$ training distribution) is $3\text{-}4\times$ higher

9 FUTURE WORK AND LIMITATIONS (1 PAGE)

While our Fine-Tuned Audio Language Model (ALM) demonstrates significant improvements in preserving character identity and emotional alignment compared to cascaded baselines, several critical limitations remain. We categorize these challenges into acoustic environment constraints, diverse emotional modeling, and architectural stability.

9.1 LIMITATIONS

Background Acoustic Contamination (The “Laugh Track” Problem) A unique challenge of using sitcom data like *Friends* is the prevalence of “canned laughter” and audience reactions. Although we applied source separation techniques during data preprocessing, residual background noise persists in the training data. Consequently, the Fine-Tuned ALM occasionally “hallucinates” faint laughter or audience applause at the end of generated utterances, even when the context is serious. This acoustic leakage compromises the purity of the character’s voice and breaks immersion during user interaction.

Difficulty in Modeling High-Context Irony and Sarcasm Our current evaluation metrics (VAD correlation and Emo2Vec) focus heavily on acoustic energy and tonal shifts. However, characters like Chandler Bing rely heavily on *sarcasm*, which often manifests as a dissonance between the text (positive content) and the audio (flat or negative prosody). Our model occasionally struggles to decouple the semantic sentiment from the acoustic delivery. When the text is overtly positive, the model tends to generate a “happy” tone, missing the “deadpan” delivery required for authentic sarcasm. This highlights a limitation in the model’s ability to perform cross-modal reasoning for complex, high-context emotions.

Long-Term Emotional Consistency Our current generation pipeline operates on a turn-level basis or with a short context window. It lacks a “Global Emotional Memory.” As a result, the model fails to maintain a consistent emotional arc across a long interaction. For instance, if Ross is “angry” in turn 1, he might abruptly switch to a “neutral” baseline in turn 2 if the text of turn 2 is not explicitly emotionally charged, ignoring the lingering effects of the previous mood state.

Inference Latency and Autoregressive Instability The ALM architecture generates audio in a discrete token space autoregressively. This process is computationally expensive and introduces significant latency, making real-time interaction difficult. Furthermore, we observed phenomena typical of codec-based language models, such as “babbling” (repeating the same phoneme indefinitely) or skipping words entirely when the probability distribution becomes too flat. While fine-tuning improved style, it did not fully resolve these stability issues inherent to the base architecture.

Data Imbalance and Rare Emotions The *Friends-MMC* dataset, while large, follows a long-tail distribution regarding emotion. “Neutral” and “Happy” samples vastly outnumber “Fear,” “Disgust,” or specific “Awkwardness.” Consequently, the model performs poorly when asked to role-play these underrepresented emotional states, often reverting to a generic prosody due to insufficient supervision signals.

9.2 FUTURE WORK

To address these limitations, we propose several directions for future research. First, for **advanced source separation and denoising**, future iterations will employ more aggressive audio-visual source separation models to isolate clean speech from laugh tracks before tokenization. As an alternative strategy, we could also introduce explicit “noise tokens” during training so the model can learn to control and suppress background sounds during inference.

Second, to **better capture sarcasm**, we propose a contrastive learning objective that penalizes the model when the generated audio emotion aligns too closely with the textual sentiment in sarcastic contexts. In other words, we aim to train the model to recognize that for certain personas, the relationship between text and audio emotion is intentionally mismatched, such that $\text{Text} \neq \text{Audio Emotion}$.

Third, we plan to explore **Global Style Tokens (GST) with memory banks** by integrating a hierarchical memory network that stores a "mood vector" updated after every turn. This would allow the model to condition its current generation not only on the immediate text, but also on the accumulated emotional state across the entire conversation session.

Finally, to address **latency and stability issues**, we could investigate a hybrid non-autoregressive refinement approach. In this setup, the ALM would first generate a coarse-grained "semantic skeleton" in the form of semantic tokens, and a faster non-autoregressive diffusion model, such as VoiceLDM, would then render the final acoustic waveform.

10 ETHICAL CONCERNS AND CONSIDERATIONS (UNINTENTIONAL, MALICIOUS, AND DUAL-USE)

Our system targets fictional characters, but it relies on techniques for high-fidelity voice and emotion cloning that could generalize to real people. Even within our controlled setting, misalignment between user intent and generated emotion (for example, sounding cheerful in a serious context) can lead to users feeling misunderstood or emotionally invalidated. Because the agent speaks in a familiar, expressive voice, users may overestimate its empathy or reliability, which raises concerns about over-attachment and over-reliance on synthetic companions.

The use of *Friends* also embeds a specific cultural and social perspective: humor, gender roles, and emotional norms are anchored in 1990s American sitcom conventions. Models fine-tuned on this material risk reproducing narrow or stereotypical patterns if later used outside this context. We therefore treat our work as a domain-specific study rather than a culturally neutral solution, and we recommend that any downstream use document data sources, cultural scope, and known biases.

At a broader level, our techniques are clearly dual-use. The same methods that enable engaging role-play could be repurposed for impersonation, harassment, or emotionally manipulative interactions if applied to real voices without consent. To mitigate these risks, we restrict our experiments to research settings, do not release general-purpose character voice models, and emphasize evaluation code and aggregate metrics rather than turnkey cloning tools. Any future deployment should combine consentful data collection, content moderation, watermarking or traceability for synthetic audio, and clear user-facing disclosures about the synthetic and fallible nature of these agents.

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A APPENDIX

You may include other additional sections here.